

CIFAR-10 Image Classification: Standard CNN vs Depthwise-Separable CNN

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Abstract

This report presents the design and implementation of image classification models on the CIFAR-10 dataset. We implement a baseline CNN with standard convolutions and a variant in which at least two standard convolution layers are replaced by depthwise convolution followed by pointwise (1×1) convolution. Both models are trained on the official training set and evaluated on the test set. We report Accuracy, Precision, Recall, F1-score, and confusion matrices for class-wise evaluation. Hyperparameter tuning is performed using Grid Search (3 hyperparameters, 3 values each) and Optuna. We compare the two architectures in terms of parameter count and computational efficiency (MACs). Results show that the Standard CNN achieves higher test accuracy (80.32%) than the Separable CNN (74.85%), while the Separable CNN uses fewer parameters and about $3.7\times$ fewer MACs, demonstrating a clear efficiency-accuracy trade-off.

1. Introduction

The assignment requires designing and implementing an image classification model on CIFAR-10 using a CNN where at least two standard convolution layers are replaced by depthwise convolution followed by pointwise (1×1) convolution, and comparing it with a regular convolution in terms of parameter count and computational efficiency. The dataset is divided into training and test sets; the model is trained on the training data and evaluated on the test set. We report performance metrics (Accuracy, Precision, Recall, F1-score) and the confusion matrix for class-wise evaluation. Hyperparameter tuning is performed using Grid Search and Optuna with three hyperparameters and three values each.

2. Dataset and Split

We use the **CIFAR-10** dataset: 60 000 colour images of size 32×32 in 10 classes (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck). The official split provides 50 000 training and 10 000 test images. The dataset is **class-balanced**: each class has exactly 6 000 images (5 000 train, 1 000 test), so no class imbalance correction is required. For hyperparameter tuning and early stopping we split the training set into 80% train and 20% validation using `random_split`. The test set is used only for final evaluation

and reporting of metrics. Data augmentation for training: `RandomCrop(32, padding=4)` and `RandomHorizontalFlip`; normalization is applied using CIFAR-10 channel means and standard deviations. The test set uses only normalization.

3. Model Architecture

3.1. Standard CNN

The baseline **Standard CNN** (`AdaptiveCNN` with `use_separable=False`) consists of:

- Conv1: $3 \rightarrow 32$ channels, 3×3 , BatchNorm, ReLU, MaxPool(2,2).
- Conv2: $32 \rightarrow 64$ channels, 3×3 (standard convolution), BatchNorm, ReLU, MaxPool(2,2).
- Conv3: $64 \rightarrow 128$ channels, 3×3 (standard convolution), BatchNorm, ReLU, MaxPool(2,2).
- FC1: $128 \times 4 \times 4 \rightarrow 512$, ReLU, Dropout.
- FC2: $512 \rightarrow 10$ (logits).

3.2. Depthwise-Separable CNN

In the **Separable CNN** (`use_separable=True`), the second and third convolutional layers are replaced by **depthwise convolution** followed by **pointwise** (1×1) **convolution**:

- **Depthwise**: `Conv2d(in_ch, in_ch, kernel_size, groups=in_ch)` — one filter per input channel.
- **Pointwise**: `Conv2d(in_ch, out_ch, kernel_size=1)` — linear combination across channels.

Thus at least two standard convolution layers (Conv2 and Conv3) are replaced by depthwise-separable blocks, as required. The rest of the architecture (Conv1, FC layers) is unchanged.

Why separable reduces MACs. For a single layer with $K \times K$ kernel, C_{in} input and C_{out} output channels (ignoring spatial dimensions), the cost in multiply-accumulates is:

- **Standard convolution**: $K \times K \times C_{\text{in}} \times C_{\text{out}}$.
- **Depthwise-separable**: depthwise $K \times K \times C_{\text{in}}$ plus pointwise $C_{\text{in}} \times C_{\text{out}}$, i.e. $K \times K \times C_{\text{in}} + C_{\text{in}} \times C_{\text{out}}$.

The ratio is $\frac{K^2 C_{\text{in}} + C_{\text{in}} C_{\text{out}}}{K^2 C_{\text{in}} C_{\text{out}}} = \frac{1}{C_{\text{out}}} + \frac{1}{K^2}$, which is much smaller than 1 when C_{out} and K^2 are moderate, hence the large reduction in MACs for the separable variant.

4. Training Setup

Both models are trained with:

- Loss: Cross-Entropy.
- Optimizer: Adam.
- Best hyperparameters (from Optuna): learning rate 0.001, batch size 128, dropout 0.2.
- Final training: 15 epochs with the chosen hyperparameters.

Hyperparameter tuning was performed only on the Standard CNN; the same best configuration is then used for both Standard and Separable CNNs so that the comparison reflects architecture alone. A fixed random seed (42) is used for reproducibility. Parameter count is computed as the sum of trainable parameters; computational cost is estimated using MACs (Multiply-Accumulate operations) via forward hooks on `Conv2d` and `Linear` layers.

5. Hyperparameter Tuning

All tuning (Grid Search and Optuna) is done on the **Standard CNN only**; the best configuration is applied unchanged to the Separable CNN for final training.

5.1. Grid Search

Grid Search is performed over **3 hyperparameters**, each with **exactly 3 values**:

- Learning rate: $\{0.0001, 0.001, 0.01\}$
- Batch size: $\{32, 64, 128\}$
- Dropout: $\{0.2, 0.3, 0.5\}$

This yields $3 \times 3 \times 3 = 27$ configurations, all of which are evaluated (the code explicitly iterates over all 27 combinations). Each configuration is trained for 5 epochs on the Standard CNN; validation accuracy is used to select the best. Best Grid Search result: `lr=0.001`, `batch_size=64`, `dropout=0.3` with validation accuracy 71.73%.

5.2. Optuna

Optuna is used with the same three hyperparameters and the same categorical choices (3 values each). We run 10 trials; each trial trains the Standard CNN for 5 epochs and returns validation accuracy. Best Optuna result: `lr=0.001`, `batch_size=128`, `dropout=0.2` with validation accuracy 71.62%. Final training uses the **Optuna-best** (rather than Grid-best) hyperparameters for both architectures: both gave similar validation accuracy, and we use the Optuna configuration for consistency with the Optuna study when training both models for 15 epochs.

6. Results

6.1. Performance Metrics

Final test-set metrics (weighted Precision, Recall, F1) and Accuracy are reported in Table 1. The Standard CNN achieves higher accuracy and F1 than the Separable CNN; both models show similar precision–recall balance.

Table 1: Test-set performance metrics (%).

Model	Accuracy	Precision	Recall	F1-Score
Standard CNN	80.32	80.77	80.32	80.31
Separable CNN	74.85	75.31	74.85	74.71

6.2. Parameter Count and Computational Efficiency

Table 2 compares parameter count and MACs. The Separable CNN has fewer parameters and substantially fewer MACs (about $3.7\times$ reduction), at the cost of lower accuracy. This

illustrates the efficiency–accuracy trade-off of depthwise-separable convolutions.

Table 2: Parameter count and MACs (efficiency comparison).

Model	Parameters	MACs
Standard CNN	1,147,914	11,375,616
Separable CNN	1,066,666	3,097,600

6.3. Training and Validation Curves

Figure 1 shows the learning curves (accuracy and loss) for the Standard CNN over 15 epochs. Figure 2 shows the same for the Separable CNN. Both models converge without severe overfitting; the Standard CNN reaches higher validation accuracy.

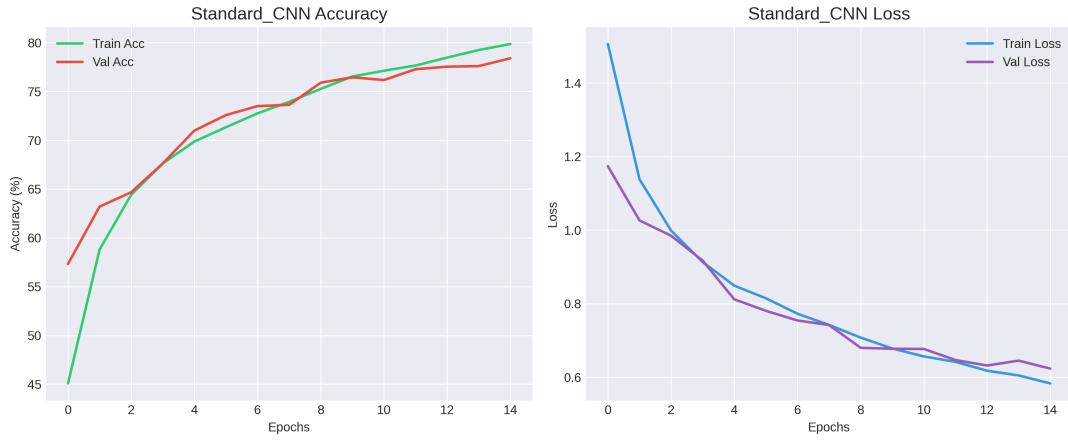


Figure 1: Standard CNN: training and validation accuracy (left), training and validation loss (right).

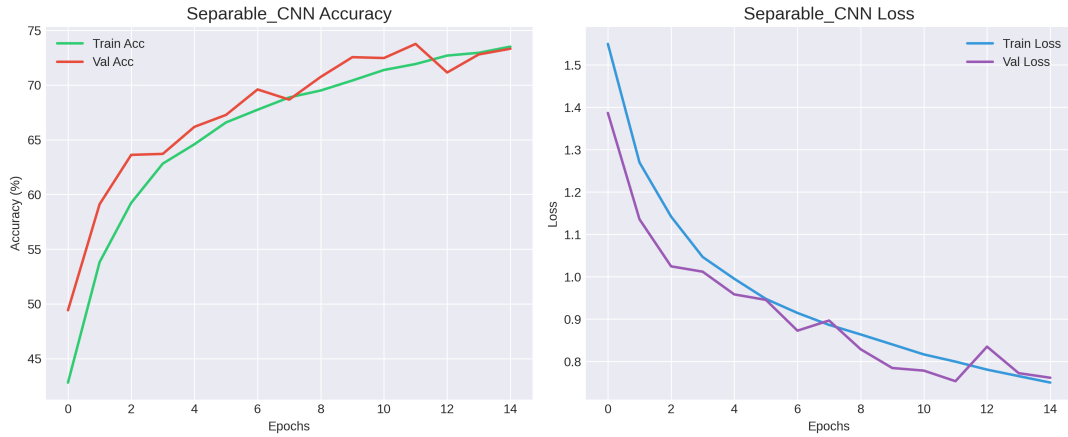


Figure 2: Separable CNN: training and validation accuracy (left), training and validation loss (right).

6.4. Confusion Matrices (Class-wise Evaluation)

The confusion matrices (Figures 3 and 4) provide class-wise evaluation. Diagonal entries indicate correct predictions per class; off-diagonal entries show confusions (e.g., cat–dog,

automobile–truck). The Standard CNN shows stronger diagonal dominance and fewer confusions than the Separable CNN.

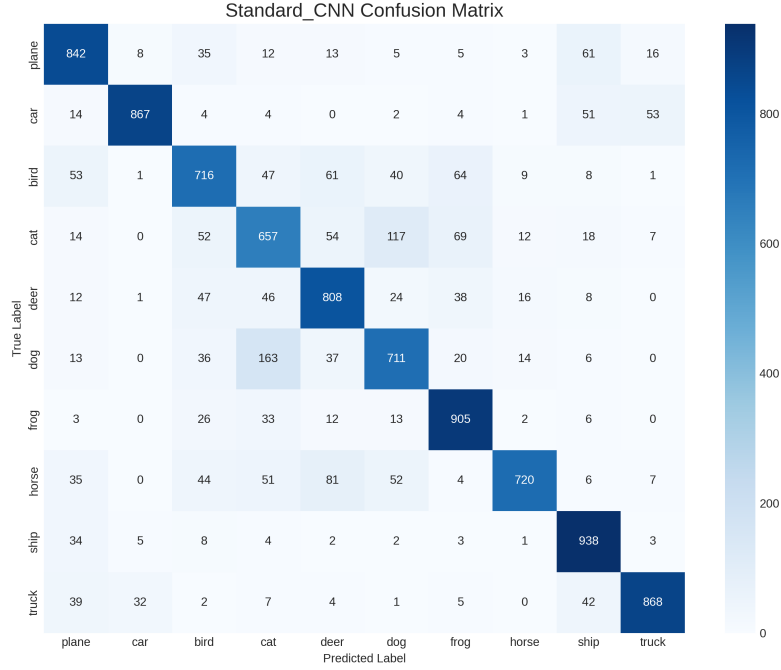


Figure 3: Standard CNN: confusion matrix on the test set (class-wise evaluation).

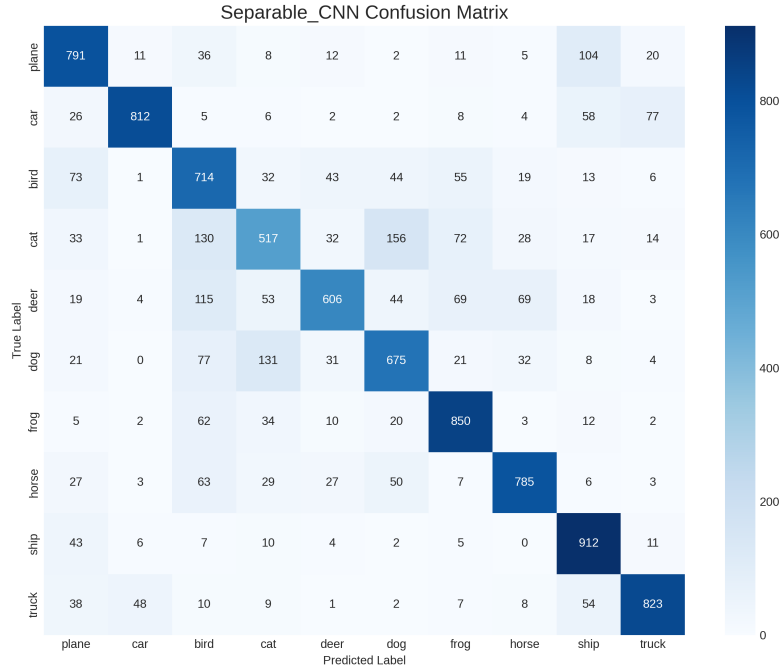


Figure 4: Separable CNN: confusion matrix on the test set (class-wise evaluation).

6.5. Efficiency Comparison (Parameters and MACs)

Figure 5 compares both models in terms of parameter count and MACs. The Separable CNN reduces both metrics, with a large reduction in MACs, making it more suitable for

resource-constrained settings where a modest drop in accuracy is acceptable.

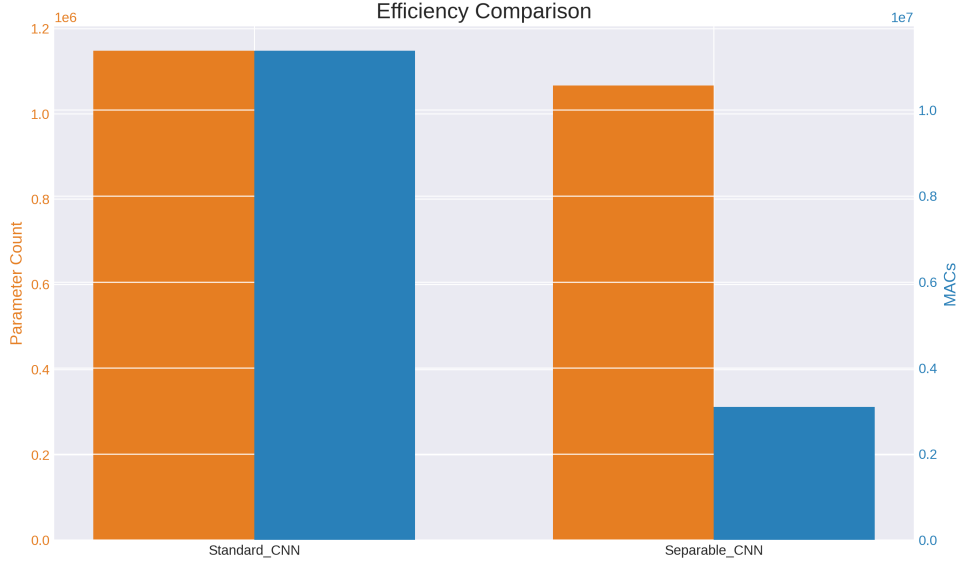


Figure 5: Efficiency comparison: parameter count (left axis) and MACs (right axis) for Standard CNN vs Separable CNN.

7. Conclusion

We designed and implemented a Standard CNN and a Depthwise-Separable CNN on CIFAR-10, with at least two standard convolution layers replaced by depthwise + pointwise convolutions in the separable variant. The dataset was used with a train/validation/test split; models were trained on the training set and evaluated on the test set. We reported Accuracy, Precision, Recall, F1-score, and confusion matrices for class-wise evaluation. Hyperparameter tuning was performed with Grid Search and Optuna (3 hyperparameters, 3 values each). The Standard CNN achieved 80.32% test accuracy with 1.15M parameters and 11.4M MACs; the Separable CNN achieved 74.85% test accuracy with 1.07M parameters and 3.1M MACs. The depthwise-separable design yields a large reduction in compute (MACs) and a modest reduction in parameters and accuracy, demonstrating the expected efficiency-accuracy trade-off.